HAORAN ZHANG

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EDUCATION

University of Texas at Austin August 2025-Present
Ph.D. in Electrical and Computer Engineering, Supervisor: Prof. Haris Vikalo

Carnegie Mellon University (CMU) May 2025
M.S. in Electrical and Computer Engineering (Advanced Study), Supervisor: Prof. Carlee Joe-Wong

Huazhong University of Science and Technology (HUST) June 2023
B.E. in Automation (Advanced Class)

The Technical University of Munich (TUM) April 2023 - August 2023
Exchange student in Electrical Engineering Department

RESEARCH INTERESTS

Optimization for (distributed) AI systems: federated learning [1]–[6]; parameter-efficient fine-tuning for LLMs [1]; online learning for hierarchical inference [7]; efficient LLM deployment [8]; medical AI [9].

PUBLICATIONS

[1] H. Zhang, D. Kim, S. Cha, and H. Vikalo, "Fedrot-lora: Mitigating rotational misalignment in federated lora," in *International Conference on Machine Learning*, 2026.
[2] M. Siew, **H. Zhang**, J.-I. Park, *et al.*, "Fair concurrent training of multiple models in federated learning," *arXiv preprint arXiv:2404.13841*, 2024.
[3] **H. Zhang**, Z. Li, Z. Gong, M. Siew, C. Joe-Wong, and R. El-Azouzi, "Poster: Optimal variance-reduced client sampling for multiple models federated learning," in *2024 IEEE 44th International Conference on Distributed Computing Systems (ICDCS)*, Best Poster Award, IEEE, 2024.
[4] Z. Gong*, **H. Zhang***, M. Siew, C. Joe-Wong, and R. El-Azouzi, "Group-based client sampling in multi-model federated learning," Under Review at ICASSP 2025 (* for equal contribution).
[5] **H. Zhang**, Z. Gong, Z. Li, M. Siew, C. Joe-Wong, and R. El-Azouzi, "Towards optimal heterogeneous client sampling in multi-model federated learning," *arXiv preprint arXiv:2504.05138*, 2025.
[6] H. Zhang, "Optimally leveraging stale updates in federated learning," *ACM SIGMETRICS Performance Evaluation Review*, vol. 53, no. 3, pp. 20–22, 2026.
[7] H. Zhang, S. Cha, H. B. Beytur, K. S. Chan, G. de Veciana, and H. Vikalo, "Online learning for multi-layer hierarchical inference under partial and policy-dependent feedback," *arXiv preprint arXiv:2603.04247*, 2026.
[8] S. Cha, H. Chen, D. Kim, *et al.*, "Regularized calibration with successive rounding for post-training quantization," *arXiv preprint arXiv:2602.05902*, 2026.
[9] **H. Zhang** and H. Chen, "Efficient 3d transformer with cluster-based domain-adversarial learning for 3d medical image segmentation," in *2023 IEEE 20th International Symposium on Biomedical Imaging (ISBI)*, IEEE, 2023, pp. 1–5.

ACHIEVEMENTS

IEEE-Eta Kappa Nu (HKN) member, Best Poster Award ICDCS 2024, Outstanding Graduate (HUST)

SKILLS

SGLang, verl, LangChain, Pytorch, Tensorflow, Sklearn, Pandas, Numpy